

Fine and ultrafine particles of the diet: influence on the mucosal immune response and association with Crohn's disease.



Crohn's disease is a modern Western disease characterised by transmural inflammation of the gastrointestinal tract. It is of unknown aetiology, but evidence suggests that it results from a combination of genetic predisposition and environmental factors. Bacterial-sized microparticles (0.1-1.0 microm) are potent adjuvants in model antigen-mediated immune responses and are increasingly associated with disease. Microparticles of TiO₂ and aluminosilicate accumulate in macrophages of human gut-associated lymphoid tissue where the earliest signs of lesions in Crohn's disease are observed. Dietary microparticles are of endogenous or exogenous origin. Endogenous microparticles dominate and are calcium phosphate (most probably hydroxyapatite), which precipitates in the lumen of the mid-distal gastrointestinal tract due to secretion of Ca and phosphate in the succus entericus. Exogenous dietary microparticles are contaminants (soil and/or dust) and food additives. TiO₂, for example, is a food colourant, and aluminosilicates are anti-caking agents, although some aluminosilicates occur as natural contaminants. Food additives alone account for ingestion of approximately 10(12) particles/person per d. Possible mechanisms for the role of exogenous and endogenous dietary microparticles in promoting toleragenic or immune responses of gastrointestinal mucosal phagocytosis are discussed. In a double-blind randomised pilot study we have shown that a diet low in Ca and exogenous microparticles appears to alleviate the symptoms of ileal Crohn's disease, with a significant ($P= 0.002$) improvement in the Crohn's disease activity index. A multi-centre trial and further mechanistic studies at the cellular level are underway.